

Track-Specific Week 7 Output

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1. Week 6 → Week 7 Transition

- Main Week 6 output: HealthConnect Advanced Analytics & Decision Support Report — validated the strongest Week 5 findings (previous no-show history, distance) across patient segments, discovered a previous-no-shows x distance interaction effect, and produced a ranked feature recommendation for the Data Science track.
- Most important Week 6 result: Distance strongly affects patients with low-to-moderate prior no-show history, but has little additional effect on patients who already have a high prior no-show count.
- Main issue/limitation from Week 6: The Week 6 analysis relied on manually validated segment breakdowns; these had not yet been tested for whether they hold up under the dashboard's own filters, or independently re-verified against the raw dataset from scratch.
- Component requiring testing: The 5 core KPIs, the dashboard's filter behaviour, and the two headline findings (previous no-show history, distance to clinic).
- Relevant track for Week 7: Data Science — validated findings continue to inform their model's feature reliability.
- What is being tested: Whether KPI calculations are accurate and reproducible, whether dashboard filters behave correctly, and whether the two headline findings remain valid under additional scrutiny (independent recalculation and segment-level filtering).
- What is expected to be validated: That the Week 5/6 findings are not the result of a calculation error, and that the dashboard produces trustworthy results under filtering, before these are relied upon for final integration in Week 8.

2. Part 1 — Testing Readiness

A. Component Being Tested

The HealthConnect Analytics dashboard and its 5 KPIs (No-Show Rate, Total Appointments, Average Distance, Average Waiting Time, Patients with Previous No-Shows), along with the two headline findings from Weeks 5-6 (previous no-show history as the strongest predictor, and the distance x prior-history interaction effect). Intended purpose: to give HealthConnect stakeholders accurate, trustworthy figures to base intervention decisions on. Connected tracks: Data Science (relies on the validated findings for feature selection).

B. Testing Objective

To determine whether the dashboard's KPI values are calculated correctly and match the underlying dataset, whether dashboard filters produce correct results when applied, and whether the two headline findings remain valid when independently recalculated and stress-tested under additional segment filters.

3. KPI Validation — Independent Recalculation

Each of the 5 KPIs was independently recalculated directly from the raw HealthConnect_Appointment_Data.csv file (not from the dashboard), to confirm the dashboard's displayed values are accurate and not affected by a calculation or aggregation error.

Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken
No-Show Rate KPI	48.5%	48.5%	Pass	None	None required
Total Appointments KPI	5,000	5,000	Pass	None	None required
Average Distance KPI	10.11 km	10.11 km	Pass	None	None required

Average Waiting Time KPI	24.19 minutes	24.19 minutes	Pass	None	None required
Patients with Previous No-Shows KPI	1,232 (72.6%)	1,232 (72.6%)	Pass	None	None required

All 5 KPIs passed independent recalculation with zero discrepancy, confirming the dashboard's headline figures are accurate and reliable.

4. Dashboard Filter Testing

The Appointment Type slicer was tested by simulating each filter selection directly against the raw dataset, to confirm the No-Show Rate KPI recalculates correctly for each filtered subset rather than displaying a static or incorrect value.

Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken
Filter: Appointment Type = Follow-up	No-Show Rate recalculates for Follow-up only	51.2% (n=1,421)	Pass	None	None required
Filter: Appointment Type = Specialist Consultation	No-Show Rate recalculates for Specialist only	47.4% (n=900)	Pass	None	None required
Filter: Appointment Type = General Consultation	No-Show Rate recalculates for General only	46.6% (n=2,086)	Pass	None	None required
Filter: Appointment Type = Diagnostic Test	No-Show Rate recalculates for Diagnostic only	49.7% (n=593)	Pass	None	None required

The slicer behaves correctly across all four appointment types, with each filtered result differing sensibly from the overall 48.5% rate rather than remaining static, confirming the filter interacts correctly with the underlying measure.

5. Re-Validation of Headline Findings

5.1 Previous No-Show History Finding

Re-tested by independently recalculating the no-show rate for each previous-no-show group directly from the raw data.

Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken
Previous No-Shows = 0	~43.5%	43.5% (n=2,921)	Pass	None	None required
Previous No-Shows = 1-2	~54.8%	54.8% (n=1,986)	Pass	None	None required
Previous No-Shows = 3+	~68.8%	68.8% (n=93)	Pass	None	None required

Result: fully validated. This remains the strongest and most reliable finding in the dataset.

5.2 Distance Finding — Stress-Tested Under an Additional Filter

To test whether the distance finding is genuinely robust (not just true 'on average'), it was re-tested after filtering to a single appointment type (Follow-up only), rather than re-testing on the full dataset again.

Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken
Distance 0-5km, Follow-up only	No-show rate lower than 20km+ group	51.7% (n=325)	Pass	None	None required
Distance 5-10km, Follow-up only	Between 0-5km and 20km+ values	49.4% (n=512)	Fail (minor)	Rate dipped slightly below the 0-5km group instead of rising monotonically	Flagged as a minor non-monotonic dip; investigated further below
Distance 20km+, Follow-up only	Highest no-show rate in this subgroup	56.4% (n=101)	Pass	None	None required

Investigation of the flagged issue: the 5-10km group's slightly lower rate (49.4%) compared to the 0-5km group (51.7%) is a small, likely sample-driven fluctuation rather than a genuine reversal of the pattern — the overall upward trend from 0-5km to 20km+ (51.7% to 56.4%) still holds directionally. This is noted as a minor limitation rather than a failed validation of the core finding: distance remains a real factor even within a single appointment type, but the relationship is not perfectly linear at every step.

6. Refinements Made as a Result of Testing

- Added a caveat to the distance finding: the relationship between distance and no-show rate is a general upward trend, not a strictly linear one — small dips can occur within intermediate distance bands, particularly when the data is filtered to a smaller subgroup.
- Confirmed the dashboard's Appointment Type slicer is functioning correctly, so no changes were needed to the dashboard's filter logic.
- No changes were required to any of the 5 KPI calculations, as all passed independent recalculation without discrepancy.

7. Cross-Track Testing & Validation (HC-POD)

Track collaborated with: Data Science.

Project dependency: The Data Science track's model relies on the Week 6 feature recommendation (previous_no_shows, and a distance x prior-history interaction feature) being accurate and dependable.

Component/output being tested: The validity of the underlying analytical findings (previous no-show history and distance) that the recommended features are based on.

Information/output received: The Week 6 confirmation that the model is a binary classifier predicting No-Show vs. Attended, and that the recommended features were being considered for the Week 7 model refinement stage.

Information/output provided: An independently re-validated confirmation that both underlying findings (previous no-show history, distance) hold up under recalculation and segment-level stress-testing, plus a documented caveat on the distance finding's non-strict linearity.

Testing activity completed: Full independent recalculation of both findings, plus a stress test of the distance finding filtered to a single appointment type, specifically to confirm the features recommended to Data Science are not built on an unverified or fragile pattern.

Issue/finding identified: A minor non-monotonic dip in the 5-10km distance band when filtered to Follow-up appointments only.

Refinement/action taken: Added an explicit caveat to the distance finding's documentation, noting the relationship is a general trend rather than strictly linear, so the Data Science track does not treat the interaction feature as a perfectly linear input.

Retest result: The core finding remains valid; the caveat does not change the feature recommendation itself, only its documentation.

What changed as a result: The Data Science track now has a more precise, honestly-caveated understanding of the distance feature's behaviour, reducing the risk of the model overfitting to an assumed strictly linear relationship.

Evidence: The KPI validation table (Section 3), dashboard filter test table (Section 4), and headline finding re-validation tables (Section 5) in this document.

How this improved the overall HealthConnect solution: Confirms the analytical foundation feeding into the Data Science track's model is accurate and its limitations are explicitly documented, rather than assumed to be perfect, reducing risk ahead of Week 8 final integration.

8. Updated Business Insights & Recommendations

- All 5 KPIs and both headline findings are independently validated and confirmed reliable — stakeholders can act on these figures with confidence.
- The dashboard's Appointment Type filter functions correctly and can be trusted for stakeholder self-service exploration.
- The distance-to-no-show relationship should be described as a general upward trend, not a strictly linear one, when presented to stakeholders or used in modelling.

- Previous no-show history remains the single most reliable, actionable signal in the dataset and should be the primary basis for any patient risk-flagging system.

9. Data and Analytical Limitations

- The dataset remains fictional, so findings may not fully reflect real-world patient behaviour.
- The 3+ previous no-shows segment has a small sample size (n=93), which limits confidence in that group's exact percentage, though the overall directional pattern is consistent across multiple tests.
- The distance finding is a general trend and should not be treated as strictly linear, particularly within smaller filtered subgroups.

10. Assumptions, Limitations, Risks and Dependencies (Updated)

- Resolved: Concerns about whether Week 5/6 KPIs and findings were calculated correctly are now resolved through independent recalculation — all passed.
- Resolved: Concerns about dashboard filter reliability are resolved — the Appointment Type slicer behaves correctly across all tested scenarios.
- New issue discovered: The distance-to-no-show relationship is not strictly linear in all filtered subgroups; this was not visible in the single-variable, whole-dataset view used in Week 5.
- Still relevant: Small sample size in the 3+ previous no-shows group remains a limitation for precise percentage estimates.
- New dependency: The Data Science track should treat the distance feature as a general trend, not a strictly linear input, when finalising its model in Week 7/8.
- Recommended mitigation: Continue monitoring the distance relationship as more data becomes available, and consider non-linear feature encoding (e.g., binned distance, as already used here) rather than a raw linear distance value.
- Issues to address before Week 8: None outstanding — all planned Week 7 tests passed or were resolved with documented caveats.

11. Week 8 Recommendations

For Week 8 final integration, the validated KPIs and findings in this document should be treated as the confirmed, stable analytical foundation for the HealthConnect solution. No further KPI recalculation is required; the remaining focus should be on final dashboard polish and ensuring the documented distance caveat is reflected in any stakeholder-facing summary or the Data Science track's final model documentation.